Learn the pairs. Freeze the order.

Training learns rules; encoding only reads them. Counts, ranks, and token IDs are distinct.

1 Learn from word counts

COUNT	BASE SYMBOLS + WORD END
× 5	c a t </w>
× 2	c a t s </w>
× 3	c a p </w>
× 1	c a p s </w>

`c + a` **5 + 2 + 3 + 1 = 11**

Wins round 1; recount after every merge.

2 Freeze the merge rules

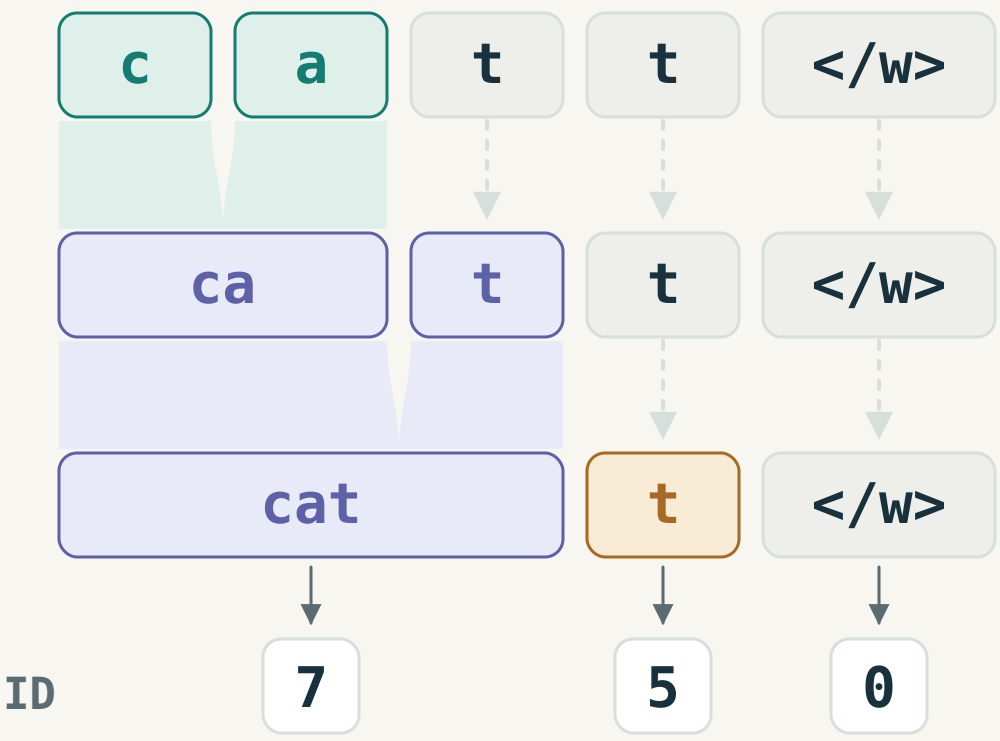
rank	FROZEN RULE	n
0	c + a → ca	11
1	ca + t → cat	7
2	cat + </w> → cat</w>	5
3	ca + p → cap	4
4	cap + </w> → cap</w>	3

n = pair count when the merge was selected

TRAINING ENDS • RULES FIXED

3 Encode the unseen word catt

Match adjacent pairs only; preserve their spans as symbols merge.



rank 0 `c + a → ca`

Merge, then recheck neighboring pairs.

rank 1 `ca + t → cat`

No recounting. No vocabulary growth.

STOP **No learned adjacent pair**

The second t blocks rank 2.

cat and </w> are not adjacent.

IDs = [7, 5, 0]. ID ≠ rank; ID 0 is the word-end marker here, not EOS.